

UCSC – Practical Take-Home Assignment QA Automation

Assignment Type: Individual Practical Assignment

Duration: 2 Weeks

Total Marks: 100

Assignment Scenario

You have joined a software company as a Junior QA Automation Engineer. Before developing automation scripts, analyse a real web application, design test scenarios, justify automation decisions, build a maintainable framework and automate selected scenarios.

Step 1 – Select a Website (5 Marks)

Choose one unique public website approved by the lecturer.

Step 2 – Requirement Analysis (10 Marks)

Analyse the application and document objectives, users, key features, assumptions, risks and observations.

Step 3 – Manual Test Scenario Design (15 Marks)

Prepare at least 15 manual test scenarios with priority, expected results and screenshots where appropriate.

Step 4 – Automation Decision (10 Marks)

Select five scenarios for automation and justify why they were chosen while explaining why the others were not.

Step 5 – Framework Design (15 Marks)

Design a Selenium + TestNG + Page Object Model framework and explain each package.

Step 6 – Automation Development (20 Marks)

Automate the selected five scenarios using Java, Selenium, TestNG and POM.

Step 7 – Debugging Challenge (10 Marks)

Create one intentional failure, identify the root cause, explain the debugging process and demonstrate the fix.

Step 8 – Reflection Video (10 Marks)

Submit a five-minute screen recording explaining your implementation and lessons learned.

Step 9 – Viva (5 Marks)

Be prepared to explain your framework and technical decisions.

Step 10 – GitHub Repository (10 Marks)

Maintain meaningful commit history throughout development.

Submission

- GitHub repository link
- Project ZIP
- PDF report
- 5-minute video
- AI_USAGE.md

Marking Scheme

Criteria	Marks
Requirement Analysis	10
Manual Test Scenarios	15
Automation Decisions	10
Framework Design	15
Automation Implementation	20
Debugging Challenge	10
Reflection Video	10
GitHub Repository	10
Total	100